Secure Energy-Efficient Scheduling for AI Workloads

Security-eligible, energy-aware placement of AI jobs on heterogeneous CPU/GPU clusters: a first reproducible prototype and a research programme

github.com/Maxxtra/phd

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Motivation

Schedulers decide where every AI job runs, yet rarely ask what a placement costs in energy, and check whether a node may *see the data* outside that decision.

- AI's energy, carbon, water and land footprints are measurable and under-reported [7, 8, 9]; deployed inference is a major share [7].
- Confidential GPUs exist, but protected execution costs throughput (3–7 % for LLM inference on H100) [6]: secure nodes are slower and scarcer.
- Energy-aware schedulers [1–3] and intra-job energy tools [4] assume every node may run every job.

Research question

With security eligibility as a hard constraint, how much *modelled* energy can placement on heterogeneous CPU/GPU clusters recover without missing more deadlines — and does security help or hurt?

H1 **seas** uses less dynamic energy than EFT *and* is non-inferior on misses (margin 2 pp).

H2 The gain grows with the share of security-constrained jobs.

H3 Both hold under ±25 % execution-time estimation error.

Preregistered (configs hashed before any run); 30 paired seeds; Holm; non-inferiority by one-sided paired t-test.

Formal model

Job j: work w_j, memory m_j, release r_j, deadline d_j, required capabilities R_j, energy weight α_j. Node n: slots, memory M_n, capabilities C_n, throughput $\theta_{n,\tau}$, powers P^{idle}_n, $P^{\text{load}}_{n,\tau}$, protected slowdown σ_n.

$$S_{jn} = \mathbb{1}\left[R_j \subseteq C_n,\ m_j \le M_n\right]$$

$$E_{jn} = (P^{\text{load}}_{n,\tau_j} - P^{\text{idle}}_n)\,\rho_{jn}\,\frac{w_j}{\theta_{n,\tau_j}}$$

$$n^* = \underset{\substack{n:\,S_{jn}=1 \\ \text{est}_{jn}+T_{jn}\le d_j}}{\arg\min}\ \alpha_j\,\hat{E}_{jn} + (1-\alpha_j)\,\hat{R}_{jn}$$

$\hat{E}$, $\hat{R}$: energy and predicted response time, normalised by the job's best option. Online, non-preemptive, least-slack-first; jobs that can no longer meet their deadline expire (same rule for all policies). Greedy heuristic; no optimality claim.

Why a node is chosen

Job j0006 · infer_small · requires TEE_CPU

deadline in 5.7 s · α = 0.5 · score = α·Ê + (1−α)·R̂

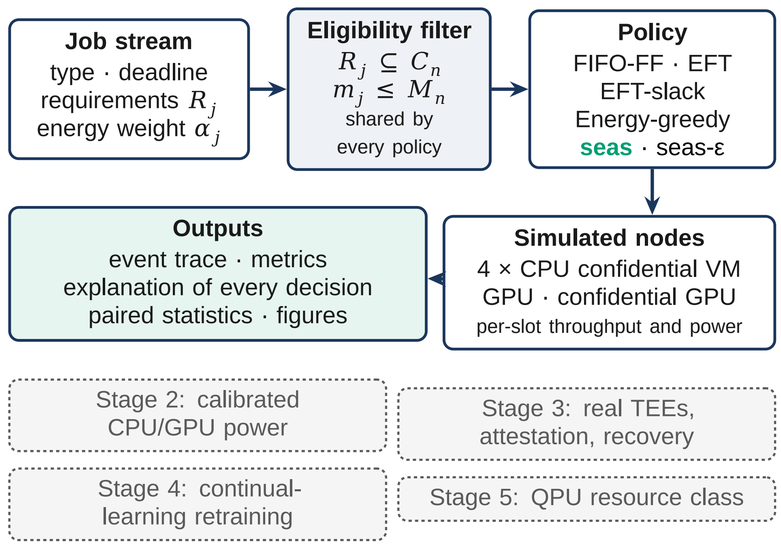
Node	est. time	est. energy	decision
gpu-0	1.1 s	136 J	rejected: lacks TEE_CPU
cpu-tee-0	25.1 s	997 J	misses deadline
cpu-tee-1	25.1 s	997 J	misses deadline
cpu-tee-2	25.1 s	997 J	misses deadline
cpu-tee-3	25.1 s	997 J	misses deadline
gpu-tee-0	1.1 s	146 J	chosen (lowest score)

A real decision record (S1, seed 0): the non-TEE GPU is fastest but ineligible; CPU confidential VMs would miss the deadline; the confidential GPU is chosen.

Limitations

- **Simulation only**: energies are modelled from synthetic profiles; no contention between co-located jobs; independent, non-preemptive jobs; one cluster.
- **Simulated security**: capability eligibility only; no enclave, encryption or attestation runs [5].

System and prototype



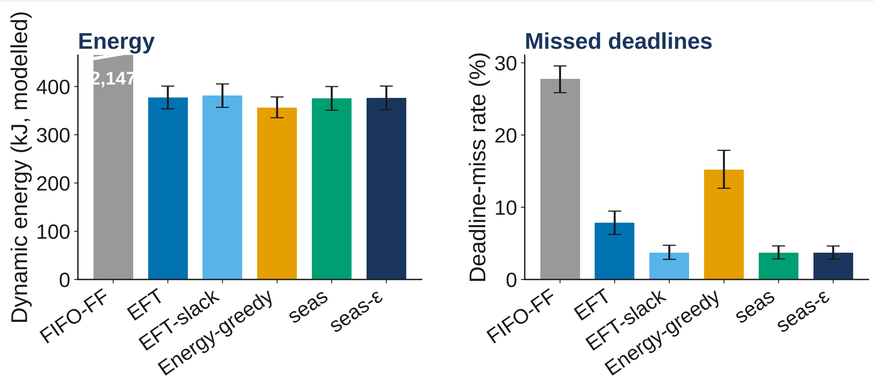
Solid: implemented and tested. Dashed: planned, not implemented.

Open-source Python package **seas**: deterministic discrete-event engine, six policies (incl. an EFT-slack ablation), explainable decision records, a paired experiment runner, statistics and verification gates. **0 eligibility violations** across 4 380 runs; every trace replayed independently against the cluster configuration.

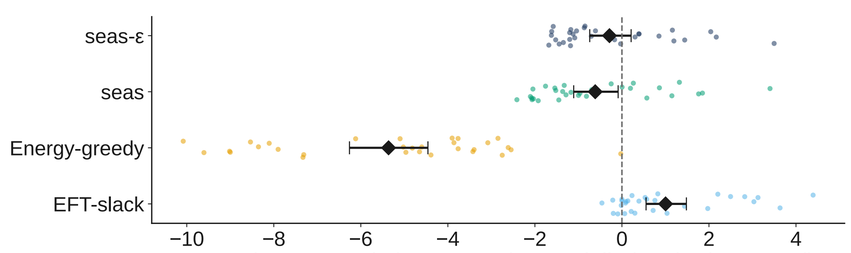
Results (S1, 30 paired seeds)

Security-eligible, deadline-aware placement halved missed deadlines (7.8 % → 3.7 %) at slightly lower dynamic energy (−0.6 %).

H1 **supported** (Holm $p = 0.040$). Energy vs EFT −0.6 % [95 % CI −1.1, −0.1]; misses non-inferior (one-sided 95 % upper bound −3.2 pp < 2 pp). Ablation: least-slack ordering alone changes energy by +1.0 %; adding the energy term changes it by −1.6 %.



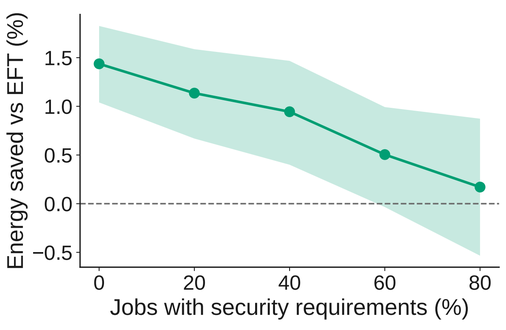
Modelled dynamic energy and deadline-miss rate per policy; mean and 95 % bootstrap CI over 30 paired seeds. FIFO-FF bar truncated (value printed).



Dynamic energy relative to EFT (%, modelled; paired per seed)

Same runs, paired per seed: dynamic energy relative to EFT. Dots = seeds, diamonds = mean with 95 % CI.

Security constraints shrink the gain (H2 refuted)



H2 **not supported**: the gain fell from 1.4 % to 0.2 % as secure jobs rose to 80 % (slope −1.6 pp per 100 % secure jobs, CI [−2.4, −0.8]). Constraints remove choices before energy can be optimised.

What the numbers mean

- **Placement leverage is small here.** Even deadline-blind energy-greedy changes energy by only −4.8 % vs **seas**, while missing 11.5 pp more deadlines (H4: deadlines binding).
- **Idle power dominates.** Dynamic energy is 18 % of the always-on total; vs EFT, total energy changes by −0.1 %.
- **Under ideal power-gating** the saving vanishes (+0.2 %, CI [−0.2, +0.6]): **seas** keeps CPUs busy longer.
- **seas** completes +3.2 % more work, so its energy per unit of completed work changes by −3.6 % [−4.1, −3.2] (secondary, not preregistered).
- Robust to estimation error (H3 supported, Holm $p = 0.002$). The energy weight acts as a switch: $\alpha \ge 0.25$ gives identical schedules.

Next question: placement must be co-optimised with power states and measured, not modelled, energy.

Research roadmap

Y1

1 · PoC (this poster) security-eligible scheduler, benchmark

2 · Hardware calibration RAPL/NVML profiles, modelled vs measured

Y2

3 · Confidential & resilient execution measured TEE overhead [5, 6], attestation, checkpointing

4 · Continual-learning-aware scheduling drift-triggered retraining jobs

Y3

5 · Quantum-classical resources QPU cost model [10], simulators first

6 · Integrated runtime each capability evaluated independently

Open research work

- **Scheduling & optimisation** — exact bounds, power-state-aware policies.
- **Energy profiling** — calibrate CPU/GPU power; validate the model.
- **Secure execution** — measure TEE overhead per job type.
- **Continual learning** — retraining streams and staleness metrics.
- **Quantum-classical** — a QPU node class in the same model.

Contributions start from a labelled first issue in the repository; reproducing every figure takes one command.

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References

[1] Wang et al., DVFS-enabled heterogeneous clusters, *IEEE TPDS* 33(12), 2022. [2] Chau et al., DVFS for CPU-GPU systems, *ACM e-Energy*, 2017. [3] Topcuoglu et al., HEFT, *IEEE TPDS* 13(3), 2002. [4] You et al., Zeus, *USENIX NSDI*, 2023. [5] Mo et al., ML with confidential computing (SoK), *ACM CSUR* 56(11), 2024. [6] Zhu et al., Confidential computing on H100, arXiv:2409.03992, 2024. [7] Luccioni et al., Power hungry processing, *ACM FAccT*, 2024. [8] Schwartz et al., Green AI, *CACM* 63(12), 2020. [9] Aczel et al., Environmental cost of AI's energy use, UNU-INWEH, 2026. [10] Beck et al., QC in HPC ecosystems, *FGCS* 161, 2024.

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